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DS 7400
HW3
Due November 9, 11:59 PM

Baseline

My HW2 model classifies 4-second windows of left-wrist accelerometry as walking, going down stairs, going up stairs, or driving. The data come from a PhysioNet dataset of 32 adults recorded at 100 Hz. I split it by person: 22 for training, 5 for validation, and 5 for testing. The network is an MLP that takes the raw window (1,200 numbers) and has two hidden layers of 64 units with ReLU and dropout (Fig. 1). I trained it with Adam and a class-weighted cross-entropy loss and kept the epoch with the best validation macro-F1. It got 89.7% test accuracy, compared with 85.2% for logistic regression.

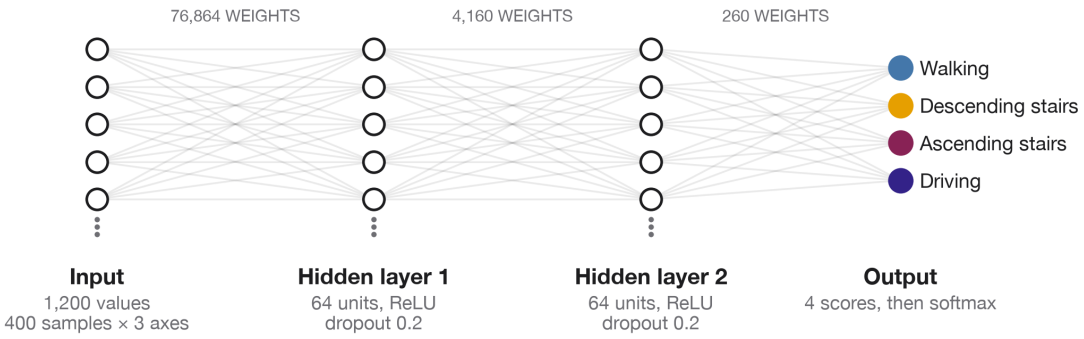


Fig. 1. The HW2 model.

Problem

The model overfits. It gets 97.5% on the training people and 87.5% on the validation people, and the validation loss goes up from the first epoch. My guess was that it learns when things happen in the window. Every input has its own weights, so a step at 1 second and a step at 2 seconds look different to it. I checked this by shifting test windows in time. The prediction changed 11.8% of the time even though the activity stayed the same.

Hypothesis

During training I shift each window by a random amount (0 to 4 seconds, with the end wrapping around to the start) and add a penalty when the prediction on the shifted window is different from the original. The penalty is the squared difference between the two sets of probabilities. I expected this to make the model focus on the rhythm of the movement and shrink the gap between training and validation accuracy. I thought it could fail if a small MLP can't learn to ignore timing, or if the model got around the penalty by predicting driving for everything.

Experiments

I compared three versions, keeping everything else the same as HW2: the HW2 model, a version trained on shifted copies without the penalty, and the version with the penalty. The middle one tells me whether a change comes from the extra shifted data or from the penalty itself. I trained each version three times with different seeds and averaged the results. I also measured how often the prediction changes when a test window is shifted.

Results

Table 1. Test results (averaged over 3 runs; ± standard deviation).

Model	Test accuracy	Macro-F1	Train − val gap	Changed after shift	Val loss (kept epoch)
HW2 model	88.9 ± 0.6%	0.60 ± 0.02	8.6 ± 1.0%	11.8 ± 0.8%	1.61 ± 0.17
Shift only	87.9 ± 0.8%	0.58 ± 0.04	10.1 ± 1.7%	8.3 ± 2.5%	0.89 ± 0.06
Shift + penalty	89.5 ± 0.8%	0.60 ± 0.02	8.1 ± 1.8%	7.0 ± 0.6%	0.87 ± 0.03

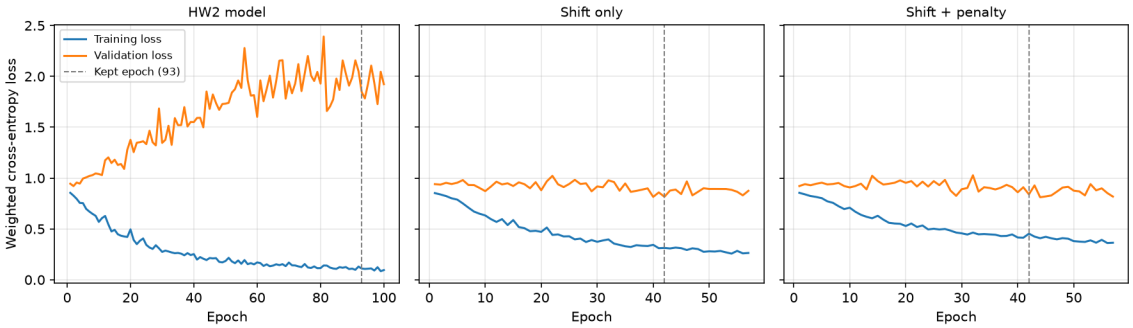


Fig. 2. Training and validation loss for one run of each version.

With the penalty, the prediction changed after a shift 7.0% of the time instead of 11.8%, and the result was about the same in all three runs. Training on shifted windows also stopped the validation loss from climbing (from 1.61 to about 0.88), with or without the penalty. Test accuracy and the gap between training and validation barely changed. The differences were about as big as the differences between runs. The model did not start predicting one class for everything. It got better at walking and going down stairs but worse at going up stairs, which also varied a lot between runs. Each training step took about twice as long because of the second forward pass.

What I learned

Part of my hypothesis held up. The model was sensitive to timing, and the penalty reduced that. Shifting also kept the validation loss from going up. It did not improve accuracy, so timing was not what was holding the model back. Most of the remaining mistakes are walking versus stairs, which may just look too similar from the wrist.